# Lab Worksheet - Exploratory Data Analysis in airquality Dataset

```r
# load libraries:
library(tidyverse) # Recall that tidyverse contains dplyr, ggplot2, and many other useful packages.

# Load `airquality` dataset:
data("airquality")
airquality |>
    head()
```

```
  Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
6    28      NA 14.9   66     5   6
```

Q1: Render this quarto file to a pdf. To do so, you can use the hotkey `Ctrl + Shift + K` or `Cmd + Shift + K`. Then, you will see a command appear in the Positron Terminal. Once the command finishes running, you will see a pdf file appear in the same directory as this file and a VIEWER window will display the pdf on the right pane of Positron. Here you should see the rendered contents of this file.

Q2: Convert the `Month` column to a factor with it's levels labeled as month names (e.g. "May", "June", etc.).

```r
# your code here
```

Q3: Create a new column `Temp_C` that converts the temperature from Fahrenheit to Celsius.

```r
# your code here
```

Q4: Rename the `Wind` column to `Wind_mph` and the Temp column to `Temp_F`.

```
# your code here
```

Q5: Create a scatter plot of `Ozone` vs `Temp_C`, colored by `Month`. Add appropriate axis labels and a title.

```
# your code here
```

Q6: Create a violin plot of `Ozone` levels for each month. Add appropriate axis labels and a title.

```
# your code here
```

Q7: Create a segmented bar chart of the proportion of days with `Ozone` levels above 50 by `Month`. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
```

Q8: Create a stacked bar chart showing the count of days available for each month. Fill bar colors by `Ozone` levels above or below 50 for each month. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
```

Q9: What is the utility of using a segmented bar chart vs a regular bar chart in this context? Which do you think is more informative and why?

# your answer here

Q10: Create a summary table of the number and proportion of NA values in each column.

```
# your code here
```